Swiggy Delivery Time Prediction Project using Traditional Machine Learning

In [ ]:

In [ ]:

In [66]:

```
#Import required libraries:
import pandas as pd
import numpy as np
import warnings
warnings.filterwarnings("ignore")
```

In [ ]:

In [67]:

```
#Load the dataset:
df = pd.read_csv("dataset/swiggy.csv")
df.head()
```

Out[67]:

	ID	Area	City	Restaurant	Price	Avg ratings	Total ratings	Food type	Address	Delivery time
0	211	Koramangala	Bangalore	Tandoor Hut	300.0	4.4	100	Biryani,Chinese,North Indian,South Indian	5Th Block	59
1	221	Koramangala	Bangalore	Tunday Kababi	300.0	4.1	100	Mughlai,Lucknowi	5Th Block	56
2	246	Jogupalya	Bangalore	Kim Lee	650.0	4.4	100	Chinese	Double Road	50
3	248	Indiranagar	Bangalore	New Punjabi Hotel	250.0	3.9	500	North Indian,Punjabi,Tandoor,Chinese	80 Feet Road	57
4	249	Indiranagar	Bangalore	Nh8	350.0	4.0	50	Rajasthani,Gujarati,North Indian,Snacks,Desser...	80 Feet Road	63

In [68]:

```
#Check for any null-values:
df.isnull().sum()
```

Out[68]:

```
ID               0
Area             0
City             0
Restaurant       0
Price            0
Avg ratings      0
Total ratings    0
Food type        0
Address          0
Delivery time    0
dtype: int64
```

In [69]:

```
#Check for any duplicate values:
df.duplicated().sum()
```

Out[69]:

```
np.int64(0)
```

In [70]:

```
df.describe()
```

Out[70]:

	ID	Price	Avg ratings	Total ratings	Delivery time
count	8680.000000	8680.000000	8680.000000	8680.000000	8680.000000
mean	244812.071429	348.444470	3.655104	156.634793	53.967051
std	158671.617188	230.940074	0.647629	391.448014	14.292335
min	211.000000	0.000000	2.000000	20.000000	20.000000
25%	72664.000000	200.000000	2.900000	50.000000	44.000000
50%	283442.000000	300.000000	3.900000	80.000000	53.000000
75%	393425.250000	400.000000	4.200000	100.000000	64.000000
max	466928.000000	2500.000000	5.000000	10000.000000	109.000000

In [71]:

```
#View basic information about the dataset:
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8680 entries, 0 to 8679
Data columns (total 10 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   ID             8680 non-null   int64
 1   Area           8680 non-null   object
 2   City           8680 non-null   object
 3   Restaurant     8680 non-null   object
 4   Price          8680 non-null   float64
 5   Avg ratings    8680 non-null   float64
 6   Total ratings  8680 non-null   int64
 7   Food type      8680 non-null   object
 8   Address        8680 non-null   object
 9   Delivery time  8680 non-null   int64
dtypes: float64(2), int64(3), object(5)
memory usage: 678.3+ KB
```

In [72]:

```
# Basic EDA:
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np  # For numeric column selection

# Set style
sns.set(style="whitegrid")

# Create 3x3 subplot grid
fig, axes = plt.subplots(3, 3, figsize=(18, 15))

# ========== ROW 1 ==========
# 1. Delivery Time Distribution
sns.histplot(df['Delivery time'], kde=True, ax=axes[0, 0])
axes[0, 0].set_title("Delivery Time Distribution")
```

```python
# 2. Price Distribution
sns.histplot(df['Price'], kde=True, ax=axes[0, 1])
axes[0, 1].set_title("Price Distribution")

# 3. Avg Ratings Distribution
sns.histplot(df['Avg ratings'], kde=True, ax=axes[0, 2])
axes[0, 2].set_title("Average Ratings Distribution")

# ========== ROW 2 ==========
# 4. Delivery Time vs Price
sns.scatterplot(x=df['Price'], y=df['Delivery time'], ax=axes[1, 0])
axes[1, 0].set_title("Price vs Delivery Time")

# 5. Delivery Time vs Avg Ratings
sns.scatterplot(x=df['Avg ratings'], y=df['Delivery time'], ax=axes[1, 1])
axes[1, 1].set_title("Ratings vs Delivery Time")

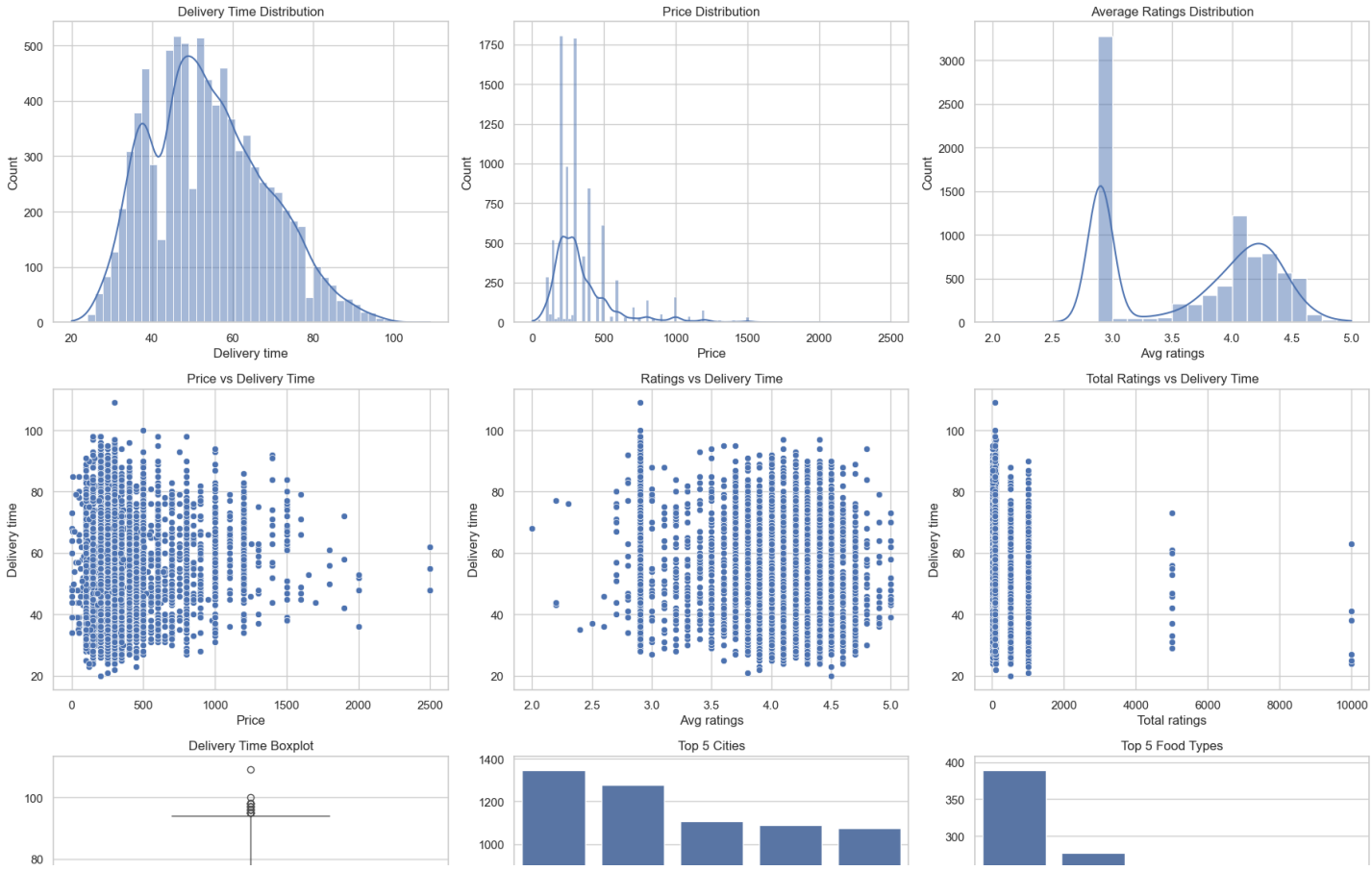
# 6. Total Ratings vs Delivery Time
sns.scatterplot(x=df['Total ratings'], y=df['Delivery time'], ax=axes[1, 2])
axes[1, 2].set_title("Total Ratings vs Delivery Time")

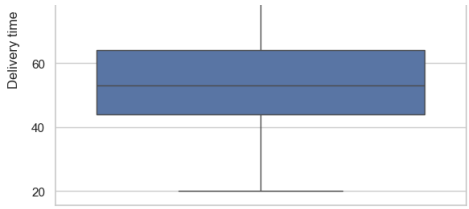
# ========== ROW 3 ==========
# 7. Boxplot of Delivery Time
sns.boxplot(y=df['Delivery time'], ax=axes[2, 0])
axes[2, 0].set_title("Delivery Time Boxplot")

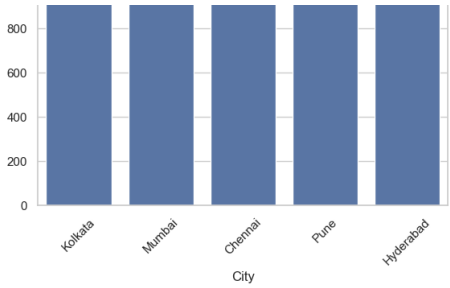
# 8. Top 5 Cities Count
top_cities = df['City'].value_counts().head(5)
sns.barplot(x=top_cities.index, y=top_cities.values, ax=axes[2, 1])
axes[2, 1].set_title("Top 5 Cities")
axes[2, 1].tick_params(axis='x', rotation=45)

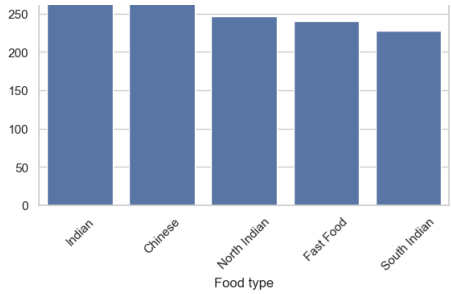
# 9. Top 5 Food Types Count
top_food_types = df['Food type'].value_counts().head(5)
sns.barplot(x=top_food_types.index, y=top_food_types.values, ax=axes[2, 2])
axes[2, 2].set_title("Top 5 Food Types")
axes[2, 2].tick_params(axis='x', rotation=45)


# Adjust layout
plt.tight_layout()
plt.show()
```









In [ ]:

In [ ]:

Q1] Predict the Delivery time of an order using restaurant, pricing, ratings, and location features.

In [73]:

```
#Feature / Target Split:
X = df.drop(columns=['Delivery time', 'ID', 'Food type'])
y = df['Delivery time']

#Train-Test Split:
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

#Preprocessing:
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import StandardScaler, OneHotEncoder

num_cols = X_train.select_dtypes(include=['int64', 'float64']).columns
cat_cols = X_train.select_dtypes(include=['object']).columns

preprocessor = ColumnTransformer([
    ('num', StandardScaler(), num_cols),
    ('cat', OneHotEncoder(handle_unknown='ignore'), cat_cols)
])
```

In [74]:

```
from sklearn.pipeline import Pipeline
from sklearn.linear_model import LinearRegression
from sklearn.neighbors import KNeighborsRegressor
from sklearn.tree import DecisionTreeRegressor
from sklearn.ensemble import ExtraTreesRegressor, AdaBoostRegressor
from xgboost import XGBRegressor


#Linear Regression:
lr_pipeline = Pipeline([
    ('preprocessing', preprocessor),
    ('model', LinearRegression())
])


#KNN Regressor:
knn_pipeline = Pipeline([
    ('preprocessing', preprocessor),
    ('model', KNeighborsRegressor())
])
```

```
#Decision-tree Regressor:
dt_pipeline = Pipeline([
    ('preprocessing', preprocessor),
    ('model', DecisionTreeRegressor(random_state=42))
])


#Extra-trees Regressor:
et_pipeline = Pipeline([
    ('preprocessing', preprocessor),
    ('model', ExtraTreesRegressor(n_estimators=250, max_depth=5, random_state=42))
])


#Adaboost:
ada_pipeline = Pipeline([
    ('preprocessing', preprocessor),
    ('model', AdaBoostRegressor(
        estimator=DecisionTreeRegressor(max_depth=7),
        n_estimators=400,
        learning_rate=0.5,
        random_state=42
    ))
])


#XG-Boost:
xgb_pipeline = Pipeline([
    ('preprocessing', preprocessor),
    ('model', XGBRegressor(
        n_estimators=500,
        learning_rate=0.05,
        max_depth=6,
        subsample=0.8,
        colsample_bytree=0.8,
        random_state=42,
        n_jobs=-1
    ))
])
```

In [75]:

```
#Model dictionary:
models = {
    "Linear Regression": lr_pipeline,
    "KNN": knn_pipeline,
    "Decision Tree": dt_pipeline,
    "Extra Trees": et_pipeline,
    "AdaBoost": ada_pipeline,
    "XGBoost":xgb_pipeline
}
```

In [76]:

```
#Cross validation:

from sklearn.model_selection import KFold, cross_validate

kf = KFold(n_splits=5, shuffle=True, random_state=42)

scoring = {
    'r2': 'r2',
    'mae': 'neg_mean_absolute_error',
    'rmse': 'neg_root_mean_squared_error'
}

results = []

for name, model in models.items():
```

```
    cv_results = cross_validate(
        model,
        X_train,
        y_train,
        cv=kf,
        scoring=scoring
    )

    mae = -np.mean(cv_results['test_mae'])
    rmse = -np.mean(cv_results['test_rmse'])
    r2 = np.mean(cv_results['test_r2'])

    results.append({
        "Model": name,
        "CV R2": round(r2, 4),
        "CV MAE": round(mae, 2),
        "CV RMSE": round(rmse, 2)
    })

results_df = pd.DataFrame(results).sort_values(by="CV R2", ascending=False)
print(results_df)
```

```
               Model   CV R2  CV MAE  CV RMSE
5            XGBoost  0.6644    6.33     8.25
1                KNN  0.6505    6.27     8.42
2      Decision Tree  0.5216    7.22     9.85
0  Linear Regression  0.4763    7.69    10.26
3        Extra Trees  0.3527    9.18    11.46
4           AdaBoost  0.3438    9.47    11.54
```

In [77]:

```
#Train the best model on Full-training data:
best_model = results_df.iloc[0]['Model']
print("Best Model:", best_model)

best_pipeline = models[best_model]
best_pipeline.fit(X_train, y_train)
```

```
Best Model: XGBoost
```

Out[77]:

```
▸ Pipeline
                                    i ?
  ▸ preprocessing: ColumnTransformer
                                    ?
  ▸ num
    ▸ StandardScaler
                                    ?
  ▸ cat
    ▸ OneHotEncoder
                                    ?
  ▸ XGBRegressor
                                    ?
```

In [78]:

```
#Test Evaluation on testing:
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

y_pred = best_pipeline.predict(X_test)

print("TEST RESULTS")
```

```
print("R2:", r2_score(y_test, y_pred))
print("MAE:", mean_absolute_error(y_test, y_pred))
print("RMSE:", np.sqrt(mean_squared_error(y_test, y_pred)))
```

```
TEST RESULTS
R2: 0.675024151802063
MAE: 6.412059307098389
RMSE: 8.241580250142036
```

In [79]:

```
#Now test on one single tuple:
single_data = pd.DataFrame({
    'Area': ['Vastrapur'],
    'City': ['Ahmedabad'],
    'Restaurant': ['Vanilla Sky'],
    'Price': [300.0],
    'Avg ratings': [2.9],
    'Address': ['Vastrapur'],
    'Total ratings': [220]
})

pred = best_pipeline.predict(single_data)
print("Predicted Delivery Time:", round(pred[0]), "minutes")
```

```
Predicted Delivery Time: 48 minutes
```

In [ ]:

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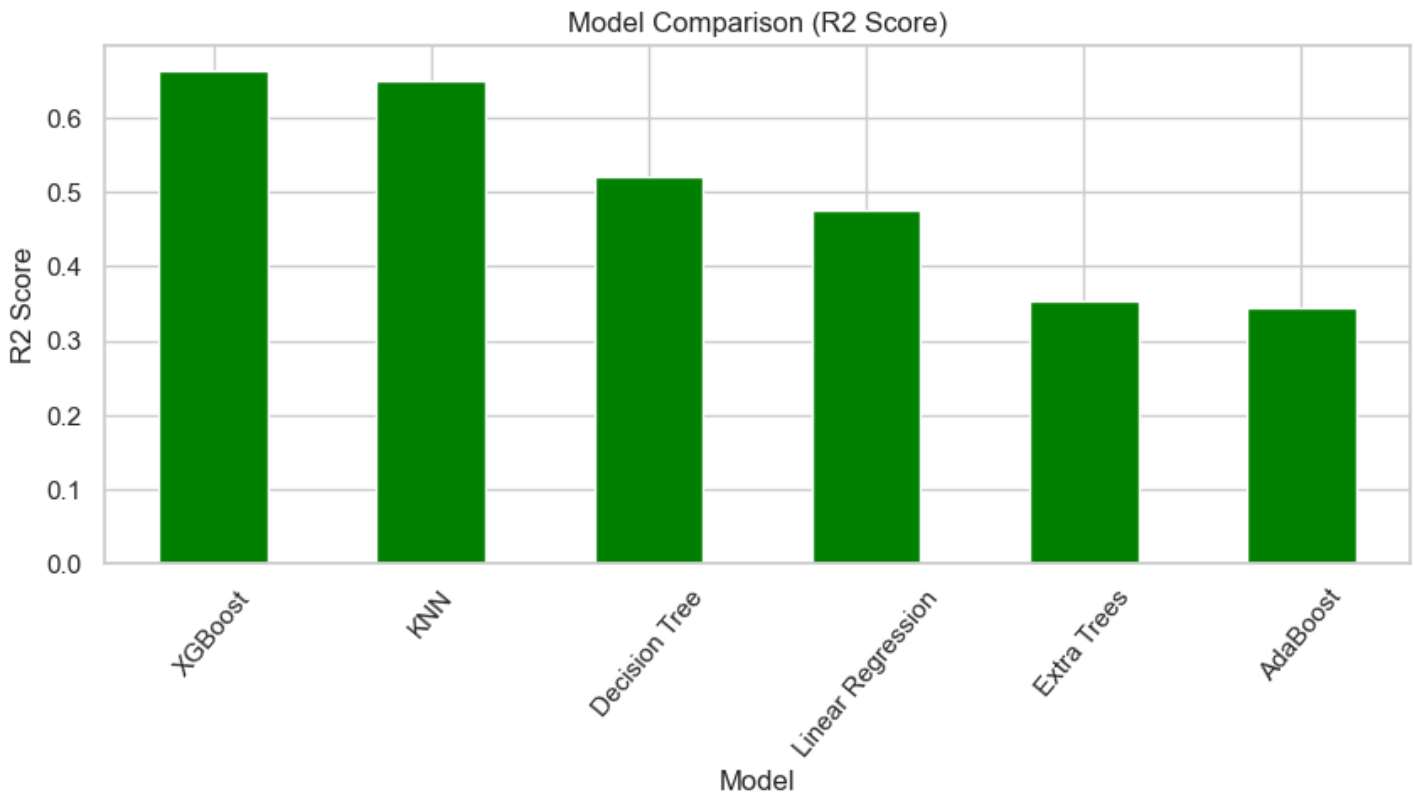
In [80]:

```
#Model comparison:
import matplotlib.pyplot as plt

plt.figure(figsize=(10,4))
results_df.set_index("Model")["CV R2"].plot(kind='bar',color="green")
plt.title("Model Comparison (R2 Score)")

plt.xlabel("Model")
plt.ylabel("R2 Score")
plt.xticks(rotation=50)
plt.show()
```



In [ ]:

In [81]:

```
#Export the best-performing model:
import joblib
joblib.dump(best_model,"Swiggy_FoodDeliveryTimePrediction_Model.pkl")
```

Out[81]:

```
['Swiggy_FoodDeliveryTimePrediction_Model.pkl']
```

In [ ]:

CONCLUSION : We used several traditional-ml algorithms to predict the delivery time. The predicted delivery time is 48 minutes.

In [ ]:

In [ ]:

Q2] Can we segment restaurants into different groups based on pricing and ratings to understand different business categories and optimize delivery strategies?

In [82]:

```
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
import matplotlib.pyplot as plt
import numpy as np

# Step 1
X = df[['Price', 'Avg ratings']]

# Step 2
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Step 3
kmeans = KMeans(n_clusters=3, random_state=42)
df['cluster'] = kmeans.fit_predict(X_scaled)

# Step 4
centroids = kmeans.cluster_centers_

# □ Step 5: Map clusters based on PRICE (column index 0)
centroid_prices = centroids[:, 0]

sorted_idx = np.argsort(centroid_prices)

labels_map = {
    sorted_idx[0]: "Low priced restaurants",
    sorted_idx[1]: "Mid priced restaurants",
    sorted_idx[2]: "Expensive restaurants"
}

# Step 6: Plot
for i in range(3):
    plt.scatter(
```

```
        X_scaled[df['cluster'] == i, 0],
        X_scaled[df['cluster'] == i, 1],
        label=labels_map[i]
    )

# Plot centroids
plt.scatter(
    centroids[:, 0], centroids[:, 1],
    color='red', marker='X', s=40, label='centroids'
)

plt.xlabel('Price (scaled)')
plt.ylabel('Avg Ratings (scaled)')
plt.title('Restaurant Segmentation using K-Means')

plt.legend()
plt.show()
```

In [83]:

```
df['cluster_label'] = df['cluster'].map(labels_map)
df['cluster_label'].value_counts().reset_index(name="Frequency of this type of restaurant
")
```

Out[83]:

	cluster_label	Frequency of this type of restaurant
0	Mid priced restaurants	4573
1	Low priced restaurants	3456
2	Expensive restaurants	651

In [ ]:

```

```

Q3] Do high-priced restaurants have significantly higher average ratings than low-priced restaurants?

In [84]:

```
high_price = df[df['Price'] > df['Price'].median()]['Avg ratings']
low_price = df[df['Price'] <= df['Price'].median()]['Avg ratings']

import numpy as np
from statsmodels.stats.weightstats import ztest

z_stat, p_value = ztest(high_price, low_price, alternative='larger')

print("Z-statistic:", z_stat)
print("P-value:", p_value)


if(p_value<0.05):
    print("High-priced restaurants have higher ratings than low-priced ones")
else:
    print("High-priced restaurants do NOT have higher ratings than low-priced ones")
```

```
Z-statistic: 9.793477790926236
P-value: 6.004630306346148e-23
High-priced restaurants have higher ratings than low-priced ones
```

In [85]:

```
from scipy.stats import ttest_ind

t_stat, p_value = ttest_ind(high_price, low_price, alternative='greater')

print("T-statistic:", t_stat)
print("P-value:", p_value)

if(p_value<0.05):
    print("High-priced restaurants have higher ratings than low-priced ones")
else:
    print("High-priced restaurants do NOT have higher ratings than low-priced ones")
```

```
T-statistic: 9.793477790926236
P-value: 7.85442528365284e-23
High-priced restaurants have higher ratings than low-priced ones
```

In [ ]:

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```

Q4] Is the average delivery time significantly different across different restaurant segments (clusters)?

In [86]:

```
from scipy.stats import f_oneway

group0 = df[df['cluster'] == 0]['Delivery time']
group1 = df[df['cluster'] == 1]['Delivery time']
group2 = df[df['cluster'] == 2]['Delivery time']

f_stat, p_value = f_oneway(group0, group1, group2)

print("F-statistic:", f_stat)
print("P-value:", p_value)


if(p_value<0.05):
    print("The average delivery time is not significantly different across different rest
aurant segments")
else:
    print("The average delivery time is significantly different across different restaura
nt segments")
```

```
F-statistic: 126.71307344603893
P-value: 5.721782939096076e-55
The average delivery time is not significantly different across different restaurant segm
```

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In [ ]:

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Q5] Can we classify whether a restaurant will have fast or slow delivery based on its price, ratings, and popularity?

In [87]:

```
df['delivery_speed'] = df['Delivery time'].apply(
    lambda x: 'Fast' if x <= df['Delivery time'].median() else 'Slow'
)


X = df[['Price', 'Avg ratings', 'Total ratings']]
y = df['delivery_speed']

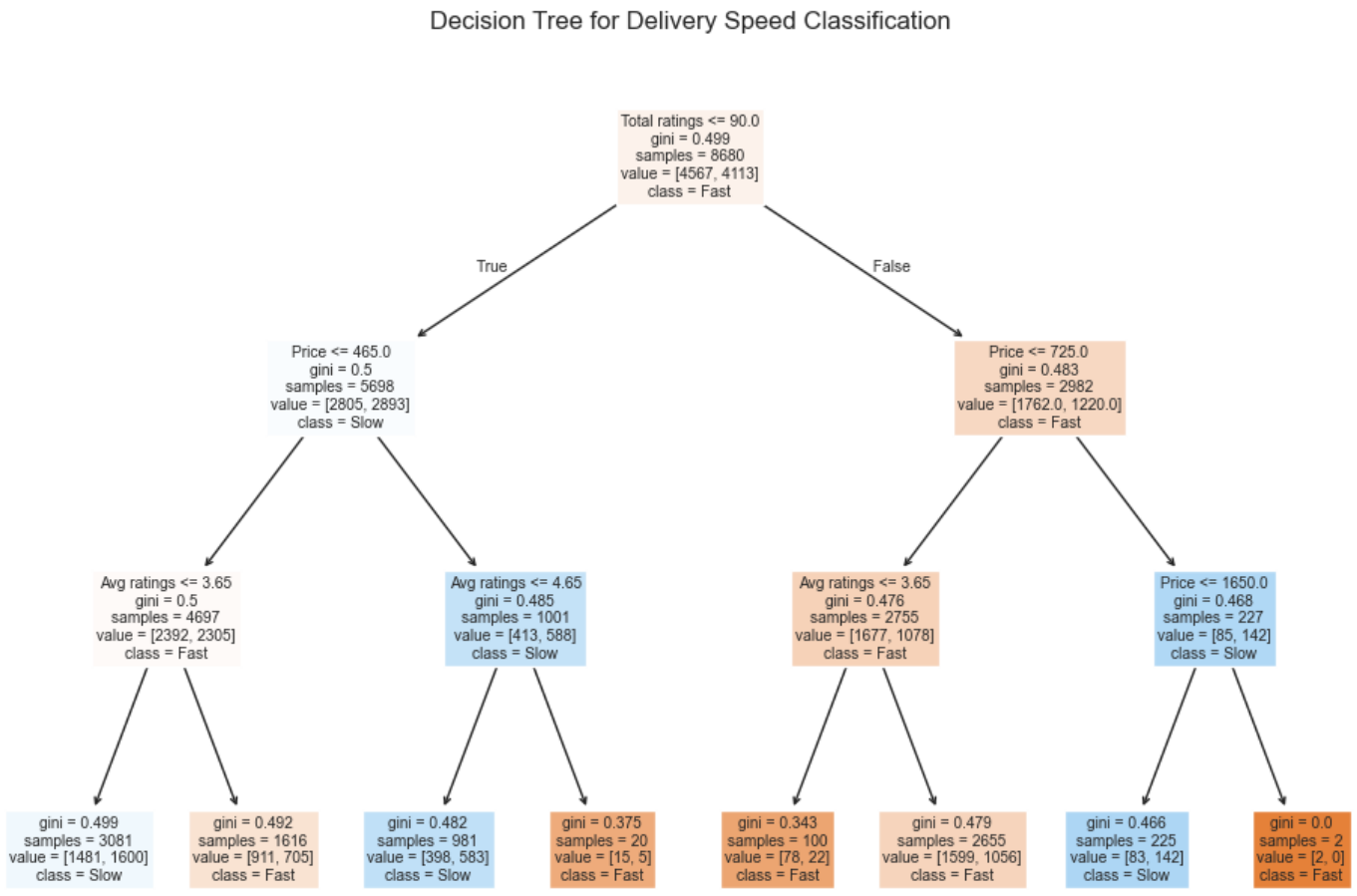

from sklearn.tree import DecisionTreeClassifier

model = DecisionTreeClassifier(max_depth=3, random_state=42)
model.fit(X, y)



from sklearn.tree import plot_tree
import matplotlib.pyplot as plt

plt.figure(figsize=(12,8))
plot_tree(
    model,
    feature_names=X.columns,
    class_names=['Fast', 'Slow'],
    filled=True
)
plt.title("Decision Tree for Delivery Speed Classification")
plt.show()
```



CONCLUSION]

- **Expensive Restaurants tend to deliver food faster.**
- **Restaurants having a more preference (based on average user ratings) tend to deliver food faster.**
- **Restaurants having low ratings (based on average user ratings) tend to deliver food slower.**

In [ ]:

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In [ ]:

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In [ ]:

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